Jess Sullivan

Agent Orchestration · Computer Vision · Full Stack Engineer · DevSecOps · ML/HPC

Lewiston, ME · 617-795-6912 · jess@sulliwood.org · github.com/jesssullivan
transscendsurvival.org · LinkedIn

Languages	ML/AI & HPC	Infrastructure
Python (10+ yrs, enterprise)	Fine-grained classification	CI/CD pipelines (expert)
TypeScript / SvelteKit (SOTA)	Horizontal parallelism in ML/AI	Kubernetes orchestration
Chapel (expert), Haskell	WASM-native inference	Binary analysis (Ghidra, Frida)
Zig, Go, C#	MLOps & model evaluation	Realtime & time-series pipelines

Experience

Full Stack Contracting and FOSS *(Ongoing)*

Ongoing contributor and community member of numerous open source projects including—and not limited to—the **Apache Foundation** (Solr web security), **rspamd**, **Chapel-lang**, **Caddy**/xcaddy, **libdns**, `Skeleton UI`, **Klipper**, **Joplin**, **FFT.js**, **KeePassXC**, `svelte-superforms`, the **Rocky Enterprise Linux Foundation** (Community Team, Kernel SIG, AltArch SIG), **Liqo** (K8s topology fabric, used by CERN), **Budgie DE**, **Mason**, along with authoring numerous FOSS automation tools, libraries and open source utilities. Expanded client list and customer references available upon request.

- Startup work: **Dover Micro** (2017), **Adaptive Motorsport** (2018); long thread of personal HID, XR, and sensor-fusion projects.
- Web GIS tools for **National Park Service**, **Foundation for Healthy Communities**, **GPRED**, **Northern Border Regional Commission**. Presented at 2019 **AAG Annual Meeting** in Washington, DC (§4).
- Fine-grained image classification with **MushroomObserver.org** and **Visipedia**; early adopter of CNN-based species ID at scale.

Current stack:

- **Web:** `SvelteKit` (Runes), `Bun`, `Vite 8` (Rolldown), TS7. Proprietary SvelteKit libraries for fingerprinting, auth, mapping, telemetry.
- **HPC:** `Chapel`, `Haskell`. Performance-oriented systems and property-based testing.
- **Tooling:** `LLVM/Clang` toolchains, `GNU Make`, `Justfile`, `Nix` flake-structured codebases.

Research:

- **Reverse Engineering & Binary Analysis:** `Ghidra`, `Frida`, `Zig` — firmware RE and NVMe controller recovery via USB bridge XRAM injection (§4). 5,400 LOC Zig tool for bypassing ASMedia ASM2362 opcode whitelists.
- **Heterogeneous Compute:** `WebGPU`, `Futhark` (GPU-targeting functional language), deeper WASM integration and WASM-native inference pipelines.
- **Functional Programming:** ESDT Monads and pixelwise classification research (pixelwise-research). `Rust` (SIMD), `Nix` (build systems).
- **Linux Kernel:** Ongoing maintainer of a Rocky 10 `PREEMPT_RT` kernel lane for scientific and HPC experimentation; upstream CVE backports against 6.1.y LTS.

Systems Analyst (DevSecOps) @ Bates College *(2024–Present)*

Scalable enterprise systems supporting staff, faculty, and ILS team. Legacy modernization; bespoke Ansible extensions/roles/plugins; 24/7 CVE mitigation; SAML and application interoperability; OpenTelemetry reporting; CI/CD pipelines (GitLab AutoDevOps, OpenTofu, RKE2 + Rancher); leading IaC adoption college-wide.

Noteworthy projects:

- Built a property-based orchestrator and instrumentation harness for degree-management and degree-auditing software in `Haskell` + `Python` (QuickCheck, Cabal, podman-compose for dev parity, FPM for packaging, AutoDevOps for CI/CD); turned an unautomatable 1980s-era C codebase unique to higher ed into a verifiable, traceable, k8s-friendly workload.
- Overhauled and automated the event-management lifecycle (`C#`, `Go`, `Ansible`).

- Led adoption of horizontally-scalable `Apache Solr` instances across multiple public and private indexing and search applications.
- Built out internal ACME-first certificate-management and DNS libraries, templates, and tooling.
- Drove enterprise secret-management practice and authentication / authorization college-wide; developed multiple SAML, LTI, Shibboleth, and bespoke TOTP / OAuth integrations; led adoption of `KeePassXC`, `firewalld`, and `fail2ban` inside declarative Ansible workflows; OTEL / LGTM / Tempo telemetry stack in production.

Fabrication Laboratory Manager @ Cornell CALS *(2021–2022)*

Developed and taught rapid fabrication curricula. `OpenSCAD`, `Fusion 360`, `Inkscape`, `C++`. GitHub/Linear project management.

Computer Vision Software Engineer @ Macaulay Library, Cornell Lab of Ornithology *(2018–2022)*

Developed & launched Merlin Sound ID, a production fine-grained audio classification system now used by millions worldwide. Led R&D on internal ML annotation tooling, model evaluation APIs, and end-to-end MLOps pipelines. Built real-time inference demos deployed at scale.

Stack:

- **Model Training:** Python (`TensorFlow`, `NumPy`, `Pandas`, `Matplotlib`, `Jupyter`)
- **Web & Inference:** `Flask`, `TypeScript`, `React`, `Vue`, `Docker`, `WebAssembly`, `React Native`, `Swift`
- **Infrastructure:** EC2, Heroku, BitBucket CI/CD, production model deployment at scale

Volunteer & Community

First Fellow @ D&M Makerspace, Plymouth State University *(2017–2020)*

Taught **Advanced GIS Programming** & **Intro to Electromechanics**. COVID-19 response: coordinated regional makerspace network for medical PPE manufacturing.

Membership Chair & 3D Printing Captain @ Ithaca Generator *(2020–2022)*

Led 501(c)(3) makerspace through rapid growth; coached hundreds via "Fusion 360 for 3D printing" series.

Business Ventures

Tinyland.dev, Inc (github.com/tinyland-inc) *(2024–present)*

Agent orchestration platform for semiautonomous infrastructure lifecycle management in higher education. Bootstrapped and in stealth; targeting source-available / zlib dual licensing.

Scope: 5 bespoke SLMs, 100+ autonomously callable tools, full IaC lifecycle coverage via a multicloud harness. Horizontal scaling with Chapel-based parallelism. Kubernetes-native architecture with Liqo-based multicloud topology.

xoxd.ai — ML/AI research arm. Bespoke SLM development, property-based testing of agent systems, and multi-cloud federation research. Chapel, Python, Go.

Previous: **Columbari.us LLC** (2017–2021) — GIS & ML contracting for UNH, NH municipal, Cornell. HPC hackerspace venture (2021–2024).

Publications

Reitsma, L.R., Burns, C., & **Sullivan, J.** (2019). *Poecile atricapillus* (Black-capped Chickadee) Feeding *Catharus guttatus* (Hermit Thrush) Nestlings. *Northeastern Naturalist*, 26(2). doi:10.1656/045.026.0213

Sullivan, J. (2026). Recovering Write-Protected NVMe SSDs Through USB Bridge XRAM Injection: Bypassing the ASMedia ASM2362 Firmware Opcode Whitelist. transscendsurvival.org/papers/recovery-paper.pdf

Presentations:

Sullivan, J. (2019). Web GIS: Telling Stories & Solving Problems. *Association of American Geographers Annual Meeting*, Washington, DC.